1 **(a)** $\frac{4}{3} \times \pi \times 3^3$ **[M1]**

36π **[A1]**

(b) $4 \times \pi \times 3^2$ **[M1]**

36π **[A1]**

[Total Marks: 4]

2 **(a)** $\pi \times 4^2 \times 9$ **[M1]**

144π **[A1]**

(b) Curved surface area $= 2 \times \pi \times 4 \times 9 = 72\pi$ **[M1]**

Area of two circular ends $= 2 \times \pi \times 4^2 = 32\pi$ **[M1]**

104π **[A1]**

[Total Marks: 5]

3 **(a)** Volume of cylinder $= \pi \times 3^2 \times 10$ $(= 90\pi)$ **[M1]**

Volume of hemisphere $= \frac{2}{3} \times \pi \times 3^3$ $(= 18\pi)$ **[M1]**

108π **[A1]**

(b) Curved surface area of cylinder $= 2 \times \pi \times 3 \times 10$ $(= 60\pi)$ **[M1]**

Area of circular base $= \pi \times 3^2$ $(= 9\pi)$ AND Curved surface area of hemisphere $= 2 \times \pi \times 3^2$ $(= 18\pi)$ **[M1]**

87π **[A1]**

[Total Marks: 6]

4 **(a)** **(i)** $\frac{120}{360} \times \pi \times 6^2$ **[M1]**

12π **[A1]**

(ii) $12\pi \times 10 = 120\pi$ **[B1]**

(b) Curved arc length $= \frac{120}{360} \times 2\pi(6) = 4\pi$ cm **[M1]**

Curved surface area $= 4\pi \times 10 = 40\pi\ cm^2$ **[M1]**

Area of two rectangular faces $= 2 \times (6 \times 10) = 120\ cm^2$ **[B1]**

Total surface area $= 120 + 40\pi + 2(12\pi) = 120 + 64\pi$ **[A1]**

[Total Marks: 7]

5 **(a)** Area of base $= \pi \times 6^2 = 36\pi$ **[M1]**

Curved surface area $= \pi \times 6 \times l = 6\pi l$ **[M1]**

$36\pi + 6\pi l = 96\pi$ leading to $6l = 60$ so $l = 10$ **[A1]**

(b) $h^2 + 6^2 = 10^2$ **[M1]**

$h = \sqrt{100 - 36} = \sqrt{64} = 8$ **[A1]**

Volume $= \frac{1}{3} \times \pi \times 6^2 \times 8$ **[M1]**

96π **[A1]**

[Total Marks: 7]

6 (a) Use of Pythagoras to find the height of the cone: $\sqrt{5^2 - 3^2}$ $(= 4)$ **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 3^2 \times 4$ $(= 12\pi)$ **[M1]**

Volume of hemisphere $= \frac{1}{2} \times \frac{4}{3} \times \pi \times 3^3$ $(= 18\pi)$ **[M1]**

30π **[A1]**

(b) Curved surface area of cone $= \pi \times 3 \times 5$ $(= 15\pi)$ **[M1]**

Surface area of hemisphere $= 2 \times \pi \times 3^2$ $(= 18\pi)$ **[M1]**

33π **[A1]**

[Total Marks: 7]

7 (a) (i) $\frac{1}{2} \times \frac{4}{3}\pi \times 6^3$ **[M1]**

144π **[A1]**

(ii) Volume of cylinder $= \pi \times 6^2 \times 5 = 180\pi$ **[M1]**

Total volume $= 144\pi + 180\pi = 324\pi$ **[A1]**

(b) $\frac{1}{3}\pi \times 9^2 \times h = 324\pi$ giving $27\pi h = 324\pi$ **[M1]**

$h = \frac{324}{27} = 12$ cm **[A1]**

(c) Slant height $l = \sqrt{9^2 + 12^2} = 15$ cm **[M1]**

Total surface area $= \pi(9)(15) + \pi(9^2) = 135\pi + 81\pi$ **[M1]**

216π **[A1]**

[Total Marks: 9]

8 (a) $\frac{1}{2} \times 6 \times 8 \times 15$ **[M1]**

360 **[A1]**

(b) Area of two triangular ends $= \frac{1}{2} \times 6 \times 8 \times 2$ $(= 48)$ **[M1]**

Area of three rectangular faces $= 15 \times (6 + 8 + 10)$ or $(15 \times 6) + (15 \times 8) + (15 \times 10)$ $(= 360)$ **[M1]**

408 **[A1]**

[Total Marks: 5]

9 **(a)** Curved surface area $= 2 \times \pi \times 5 \times 12$ **[M1]**

Total surface area $= 2 \times \pi \times 5 \times 12 + 2 \times \pi \times 5^2$ **[M1]**

534 or 534.07... or 170π **[A1]**

(b) Volume of cylinder $= \pi \times 5^2 \times 12 = 300\pi$ (942...) **[M1]**

$\frac{4}{3}\pi R^3 = 300\pi$ **[M1]**

$R = \sqrt[3]{225} = 6.08$ or 6.082... **[A1]**

[Total Marks: 6]

10 **(a)** $\frac{1}{3} \times 10^2 \times h = 400$ **[M1]**

12 **[A1]**

(b) Distance from centre of base to midpoint of side = 5 cm **[B1]**

Slant height of a triangular face $= \sqrt{12^2 + 5^2}$ $(= 13)$ **[M1]**

Area of one triangular face $= \frac{1}{2} \times 10 \times 13$ $(= 65)$ **[M1]**

360 **[A1]**

[Total Marks: 6]

11 **(a)** Calculates cross-sectional sector area: $\frac{110}{360} \times \pi \times 6^2 = 11\pi$ $(= 34.557...)$ **[M1]**

Multiplies by length: $11\pi \times 15$ **[M1]**

518 (accept 518.3 to 518.4) **[A1]**

(b) Calculates curved arc length: $\frac{110}{360} \times 2 \times \pi \times 6 = \frac{11\pi}{3}$ $(= 11.519...)$ **[M1]**

Sums all five faces: $2(11\pi) + 15 \times \left(\frac{11\pi}{3} + 6 + 6\right)$ **[M1]**

422 (accept 421.9 to 422.0) **[A1]**

[Total Marks: 6]

12 **(a)** Height of cone $= 24 - 12 = 12$ cm **[B1]**

Volume of cylinder $= \pi \times 5^2 \times 12$ $(=300\pi$ or $942.4\ldots)$ **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 5^2 \times 12$ $(=100\pi$ or $314.1\ldots)$ **[M1]**

1260 (or 1256 to 1257 or 400π) **[A1]**

(b) Slant height of cone $(l) = \sqrt{5^2 + 12^2} = 13$ **[M1]**

Curved surface area of cone $= \pi \times 5 \times 13$ $(=65\pi$ or $204.2\ldots)$ **[M1]**

Curved surface area of cylinder $= 2 \times \pi \times 5 \times 12$ ($=120\pi$ or $376.9\ldots$) **[M1]**

Area of circular base $= \pi \times 5^2$ ($=25\pi$ or $78.5\ldots$) **[M1]**

660 (or $659.7\ldots$ or 210π) **[A1]**

[Total Marks: 9]

13 (a) $\pi \times (x)^2 \times 3x$ **[M1]**

$3\pi x^3$ **[A1]**

(b) $\frac{4}{3} \times \pi \times \left(\frac{x}{2}\right)^3$ **[M1]**

$\frac{\pi x^3}{6}$ **[A1]**

(c) 18 **[B1]**

(d) Total surface area $= 2\pi x^2 + 2\pi x(3x)$ **[M1]**

$8\pi x^2 = 140$ **[M1]**

$x = 2.36$ (or $2.360\ldots$) **[A1]**

[Total Marks: 8]

14 (a) Volume of cone $= \frac{1}{3}\pi r^2(2r) = \frac{2}{3}\pi r^3$ **[B1]**

Volume of hemisphere $= \frac{1}{2} \times \frac{4}{3}\pi r^3 = \frac{2}{3}\pi r^3$ **[B1]**

Equates total volume: $\frac{2}{3}\pi r^3 + \frac{2}{3}\pi r^3 = 450\pi$ leading to $\frac{4}{3}r^3 = 450$ **[M1]**

Solves for r: $r^3 = 337.5 = 27 \times 12.5$, therefore $r = 3\sqrt[3]{12.5}$ **[A1]**

(b) Finds slant height of cone: $l = \sqrt{r^2 + (2r)^2} = r\sqrt{5}$ **[M1]**

Sets up total surface area expression: $\pi rl + 2\pi r^2 = \pi r^2(\sqrt{5} + 2)$ **[M1]**

Evaluates $r = 3\sqrt[3]{12.5} = 6.96238...$ and substitutes into area formula **[M1]**

645 (accept 645.0 to 645.4) **[A1]**

[Total Marks: 8]

15 (a) Area of trapezium $= \frac{1}{2} \times (15 + 25) \times 18$ **[M1]**

360 **[A1]**

Volume $= 360 \times 120 = 43200$ cm^3 **[M1]**

43.2 **[A1]**

(b) Volume of water after 30 seconds $= 0.8 \times 30 = 24$ litres $= 24000$ cm^3 **[M1]**

Let depth be h and the width of water surface be w.
$w = 15 + \frac{10}{18}h = 15 + \frac{5}{9}h$ **[M2]**

Area of cross-section of water $= \frac{1}{2} \times (15 + 15 + \frac{5}{9}h) \times h = 15h + \frac{5}{18}h^2$ **[M1]**

Volume $= (15h + \frac{5}{18}h^2) \times 120 = 24000$

$1800h + \frac{100}{3}h^2 = 24000$

$h^2 + 54h - 720 = 0$ **[M1]**

$(h - 11.23...)(h + 65.23...) = 0$

$h = 11.2$ or $11.23...$ **[A1]**

[Total Marks: 9]

16 (a) Volume of large cone $= \frac{1}{3} \times \pi \times 10^2 \times 24$ $(= 800\pi)$ **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 5^2 \times 12$ $(= 100\pi)$ **[M1]**

2200 or 2199.1 to 2199.2 **[A1]**

(b) Slant height of large cone $= \sqrt{10^2 + 24^2}$ $(= 26)$ AND Slant height of small cone $= \sqrt{5^2 + 12^2}$ $(= 13)$ **[M1]**

Curved surface area of frustum $= \pi \times 10 \times 26 - \pi \times 5 \times 13$ $(= 195\pi)$ **[M1]**

Total surface area $= 195\pi + \pi \times 10^2 + \pi \times 5^2$ **[M1]**

1010 or $1005.3\ldots$ **[A1]**

(c) Volume of one sphere $= \frac{4}{3} \times \pi \times 3^3$ $(= 36\pi$ or $113.1\ldots)$ **[M1]**

$\frac{700\pi}{36\pi}$ or $\frac{2199.1}{113.1}$ **[M1]**

19 **[A1]**

[Total Marks: 10]

17 (a) $\frac{1}{3} \times 12^2 \times 10$ **[M1]**

480 **[A1]**

(b) Use of Pythagoras to find slant height: $\sqrt{10^2 + 6^2}$ **[M1]**

$\sqrt{136}$ or 11.6... or 11.7 **[A1]**

Area of four triangular faces $= 4 \times \frac{1}{2} \times 12 \times$ their $\sqrt{136}$ $(= 279.8...)$ **[M1]**

424 or 423.8... to 423.9 **[A1]**

(c) Volume of one sphere $= \frac{4}{3} \times \pi \times 2.5^3$ $(= 65.4...)$ **[M1]**

$\frac{480}{\text{their } 65.4...}$ **[M1]**

7 **[A1]**

(d) $\frac{1620}{480}$ or 3.375 **[M1]**

$\sqrt[3]{3.375}$ or 1.5 **[M1]**

15 **[A1]**

[Total Marks: 12]

18 **(a)** $\frac{1}{3} \times 16^2 \times 15$ **[M1]**

1280 **[A1]**

(b) Slant height of triangular face $= \sqrt{15^2 + 8^2} = 17$ cm **[M1]**

Total surface area $= 16^2 + 4 \times \left(\frac{1}{2} \times 16 \times 17\right) = 256 + 544$ **[M1]**

800 **[A1]**

(c) **(i)** Height of removed pyramid $= 15 - 9 = 6$ cm, giving linear scale factor $= \frac{6}{15} = 0.4$ **[M1]**

Volume of frustum $= 1280 - 0.4^3 \times 1280 = 1280 - 81.92$ **[M1]**

1198.08 (or 1200) **[A1]**

(ii) Top square side $= 0.4 \times 16 = 6.4$ cm and trapezoidal slant height $= 0.6 \times 17 = 10.2$ cm **[M1]**

Total surface area $= 16^2 + 6.4^2 + 4 \times \frac{1}{2}(16 + 6.4) \times 10.2 = 256 + 40.96 + 456.96$ **[M1]**

753.92 (or 754) **[A1]**

(d) $OA = \sqrt{8^2 + 8^2} = \sqrt{128}$ $(= 11.313\ldots)$ cm **[M1]**

$\tan \angle VAO = \frac{15}{\sqrt{128}}$ **[M1]**

53.0 **[A1]**

[Total Marks: 14]